



ELSEVIER

Forest Ecology and Management 153 (2001) 89–104

Forest Ecology  
and  
Management

www.elsevier.com/locate/foreco

# Evaluating the effects of ecosystem management alternatives on elk, mule deer, and white-tailed deer in the interior Columbia River basin, USA

John F. Lehmkuhl<sup>a,\*</sup>, John G. Kie<sup>b</sup>, Louis C. Bender<sup>c,1</sup>,  
Gregg Servheen<sup>d</sup>, Harvey Nyberg<sup>e</sup>

<sup>a</sup>US Forest Service, Pacific Northwest Research Station, 1133 N. Western Ave., Wenatchee, WA 98801, USA

<sup>b</sup>US Forest Service, Pacific Northwest Research Station, 1401 Gekeler Lane, La Grande, OR 97850, USA

<sup>c</sup>Washington Department of Fish and Wildlife, 2950 Ley Rd., Cle Elum, WA 98922, USA

<sup>d</sup>Idaho Department of Fish and Game, 1540 Warner Avenue, Lewiston, ID 83501, USA

<sup>e</sup>Montana Department of Fish, Wildlife, and Parks, 2300 Lake Elmo Drive, Billings, MT 59105, USA

## Abstract

Elk (*Cervus elaphus*), mule deer (*Odocoileus hemionus*), and white-tailed deer (*Odocoileus virginianus*) are highly valued for their game, aesthetic, and spiritual qualities by sportsman, wildlife enthusiasts, and Native Americans in North America. As part of the Interior Columbia Basin Ecosystem Management Project (ICBEMP) of the US Forest Service and Bureau of Land Management, we: (1) defined key habitat associations of those species that could be used for a broad-scale (58 million hectares) analysis of management practices and (2) determined how three ecosystem management alternatives of a supplemental draft environmental impact statement (SDEIS) might affect the regional distribution of habitat for those species across the Basin over the next 100 years. For the three species, we developed a Bayesian Belief Network model that used available SDEIS datasets to estimate historical, current, and future habitat capability under the management alternatives in each of the 7467 subwatersheds (mean 8000 ha) in the study area. The model quantified “inherent habitat capability” as a function primarily of forage habitat capability, with cover as a minor influence. Forage habitat capability was a function of the percentage area of rangeland and early seral forest community types, and the qualitative influences of livestock overgrazing, wildfire, and prescribed fire. For the current and future periods, an “adjusted habitat capability” was estimated by adjusting inherent habitat capability for the negative effects of poor security from human disturbance. Open road density, cover area, and a terrain relief index were used to estimate the security effect. Habitat capability was reported by 15 ecological regions within the study area as mean subwatershed capability. Under all management alternatives, habitat capability increased about 5% for all three ungulate species over the next 100 years. Limitations of the coarse analysis scale restrict application of the model to large-scale assessments. Lacking regional population data, verification of model output was not feasible at the scale of analysis. However, the model was considered useful for tracking regional changes given the available habitat data and regional-scale objectives of the effort. Published by Elsevier Science B.V.

**Keywords:** *Odocoileus hemionus*; *Odocoileus virginianus*; *Cervus elaphus*; Forest management; Ecosystem management; Bayesian model

\* Corresponding author. Tel.: +1-509-662-4315, ext: 235; fax: +1-509-664-2742.

E-mail address: jlehmkuhl@fs.fed.us (J.F. Lehmkuhl).

<sup>1</sup> Present address: Cooperative Fish and Wildlife Research Unit, P.O. Box 30003, MSC 4901, Las Cruces, NM 88003, USA.

## 1. Introduction

Management of federally administered wildlands outside of national parks in the United States has changed during the last decade from multiple-use management emphasizing resource extraction to ecosystem management emphasizing biodiversity conservation and compatible human uses (Dombeck, 1996; Thomas and Dombeck, 1996). Scientific analysis to develop and evaluate large-scale ecosystem management strategies for wildlife management has typically focused on the population viability of plant and animal species of conservation concern (FEMAT, 1993; Lehmkuhl et al., 1997). Although traditional forest management plans on the smaller-scale of National Forests (Bettinger et al., 1996) have included wildlife of social and economic importance, e.g., game species, large-scale regional ecosystem plans have not considered those species because most are of minor viability concern. Yet, many of those species have high social or economic value in terms of recreation, aesthetics, education, or commerce (Steinhoff, 1978). Hence, they should be important components of large-scale ecosystem management strategies that attempt to meld ecological and human needs and values in ecosystem management.

Elk, mule deer, and white-tailed deer habitat and population management in the interior Columbia River basin study area (hereafter referred to as Basin) are examples of such concern. The Basin, between the crest of the Cascade Range in Oregon and Washington and the Continental Divide in Montana, Idaho, and Wyoming, is about 58 million hectares, of which about one-half is administered by the US Department of Agriculture, Forest Service (FS) and US Department of Interior, Bureau of Land Management (BLM). Management alternatives proposed for those federal lands by the Interior Columbia Basin Ecosystem Management Project (hereafter, Project) have the potential for integrating conservation, social, and economic needs to manage forest and rangeland ecosystems at multiple scales — from stands to landscapes to regions (Haynes et al., 2001).

Sportsmen, naturalists, sporting and service businesses, and state wildlife management agencies are concerned that ecosystem management might affect elk and deer habitats, populations, hunting opportunities, and the associated revenue generated by

hunters. Hunting, in general, contributes about \$146 million to local economies and supports 49 000 jobs in the Basin (Haynes and Horne, 1997). Moreover, much of this revenue benefits rural communities that have low economic resilience and are hard hit by changes in resource extraction industries sometimes associated with changing priorities for land management (McCool et al., 1997). As the land management focus changes from extraction to recreation, aesthetics, and education, it will be important to know how new land management strategies affect resources critical for rural community survival. Other concerns come from Native Americans who have deep cultural affinities with plants and wildlife (McCool et al., 1997). Elk, in particular, have been important historically to the economy and culture of Native Americans in North America, particularly tribes in the Basin, providing vital sources of food, clothing, implements, weapons, decoration, spiritualism, and a medium of exchange (McCabe, 1982).

Land management can alter the capability of the land to provide habitat for elk and deer by changing the types, quantity, and arrangement of habitats over space and time (Lyon and Ward, 1982; Thomas et al., 1979). Management also affects the quality of the habitat through manipulation of vegetation with fire (Jourdonnais and Bedunah, 1990) and grazing (Nelson, 1982) and by affecting the security with which animals can fully use habitat without disturbance. Forest and herbaceous cover, road density, and human activity associated with roads, along with the complexity of the terrain, define the amount of security from human disturbance and use of habitats by animals (Edge and Marcum, 1991; Hillis et al., 1991).

We evaluated the effects of three management alternatives (S1, S2, S3) developed by the Project for federal lands in the Basin on elk, mule deer and white-tailed deer habitat capability. Our goals were to: (1) summarize key habitat associations in a model for a broad-scale Basin-wide analysis and (2) determine how three ecosystem management alternatives in a supplemental draft environmental impact statement (hereafter, SDEIS) might affect elk and deer habitat capability across the Basin in the future. To meet those goals, we focused on the following questions:

1. What are the key broad-scale habitat variables (e.g., composition and structure of vegetation,

road-associated factors such as road density and use) controlled or influenced by BLM and FS management?

2. What are the appropriate geographic areas in which to evaluate habitat trends?
3. What are the key geographic areas of most interest and use to tribal nations?
4. What are the historical and current conditions of habitat capability across the Basin, and how will the SDEIS alternatives affect habitat trends in the short (10 years) and long (100 years) terms?

## 2. Methods

### 2.1. Ecosystem management alternatives

USDA and USDI (2000) provide a detailed discussion of the SDEIS alternatives: a brief description follows. Alternative S1 continues current management direction under existing National Forest and BLM land use plans. Many of the plans were based on the assumptions that ecological conditions are healthy and that disturbances (fire, insect and disease) would not substantially affect desired outcomes. In general, the intent is to provide sustainable levels of products (e.g., wood, forage, game, minerals) in an environmentally prudent manner in areas not set aside for wilderness or other non-consumptive uses.

Alternatives S2 and S3 focus on restoring and maintaining ecosystem structure and function, and providing for needs of people, while reducing short- and long-term risks from human and natural disturbances. Alternative S2 emphasizes analyses at sub-basin and watershed scales prior to conducting management activities to ensure actions occur in the most appropriate locations in the most appropriate sequence. Restoration is prioritized based on risk to terrestrial and aquatic habitats by unnaturally severe disturbance and the opportunity to address those risks. Alternative S3 has a focus similar to Alternative S2, but with considerably less emphasis on prior analysis. Managers will take some short-term risks to address long-term risks of disturbance events, if the risks are acceptable within the law (e.g., Endangered Species Act, Clean Air Act). Management activities are linked to areas where they can benefit isolated communities

that are economically specialized in industries tied to goods and services from federal lands (USDA and USDI, 2000).

### 2.2. Model development

We assembled an analysis team of 10 ungulate scientists and managers from state and federal government agencies and tribal governments to identify key broad-scale habitat variables and appropriate geographic analysis units, and to develop the analytical habitat model. The team initially identified key variables for the model using information from an ungulate assessment report to the Project science team (Christensen et al., 1995) and the literature. Variables were selected for modeling if they could be reliably quantified either directly or by proxy variables in the datasets available to the Project for the SDEIS analysis. Those datasets were the result of an assessment of current conditions (Hann et al., 1997), modeling to predict historical vegetation conditions (Hann et al., 1997), and modeling the future effects of the three alternatives over 10-year and 100-year periods (Hemstrom et al., 2000).

Subwatersheds of sixth hydrologic unit code (hereafter, subwatersheds) magnitude, averaging about 8000 ha, were the smallest analysis units for estimating habitat capability. The 7467 subwatersheds in the analysis area were the smallest common unit of analysis for all disciplines analyzing the SDEIS. Subwatersheds were considered an appropriate sample unit that corresponded to the scale of ungulate management and herd units. Results from individual subwatersheds were mapped and averaged by Regional Advisory Committee/Provincial Advisory Committee (RAC/PAC) geographic regions (hereafter, Regions) developed for the SDEIS analysis (Fig. 1). The 15 Regions were designed to encompass geographic areas with similar ecosystem, management, social, and economic attributes. They ranged in area from 200 000 to 6 000 000 ha. Through consultation with tribal representatives on our analysis team (Don Gentry and Craig Bienz, Klamath Tribes, Chiloquin, OR; Scott McCorquodale, Yakama Nation, Toppenish, WA) and the tribal liaison with the Project planning team (Cliff Walker, ICBEMP, Boise, ID), we determined Regions also were the key geographic areas of most interest and use to tribal nations.

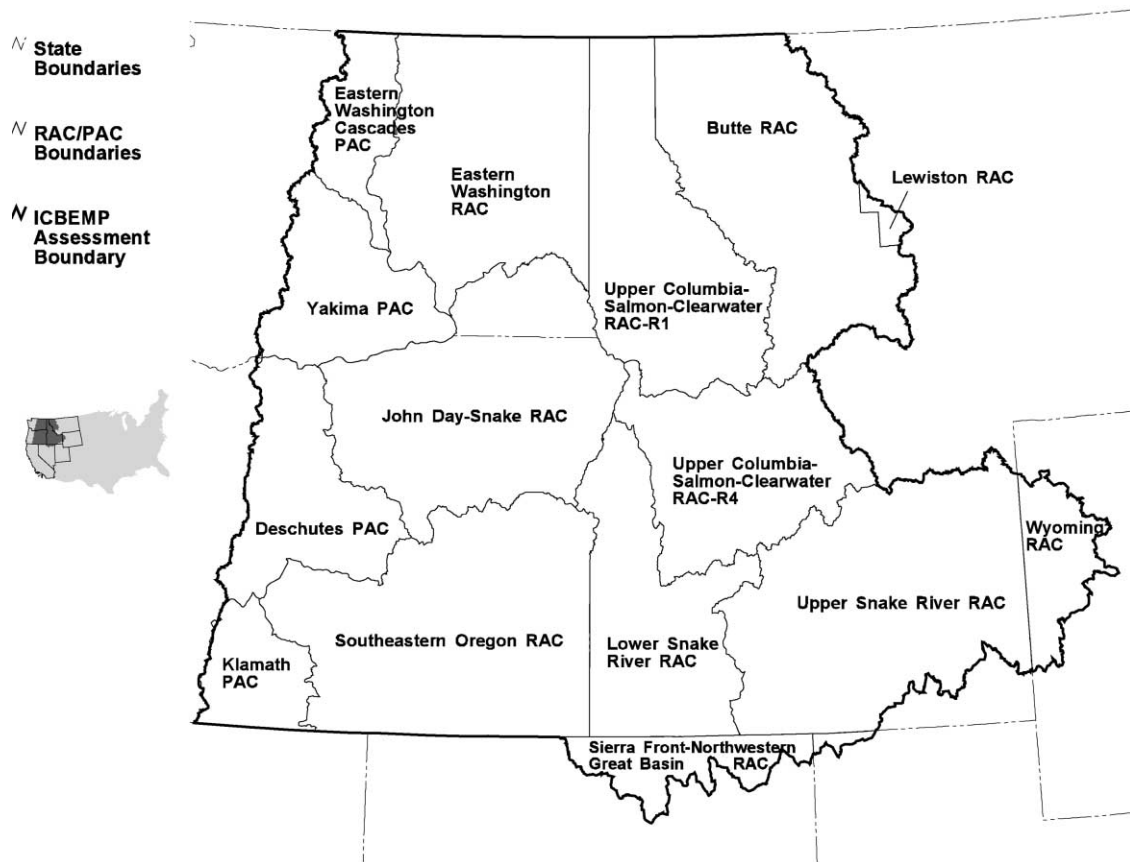


Fig. 1. Interior Columbia River basin study area in the Pacific northwest of the United States, showing Resource Advisory Council (RAC) and Provincial Advisory Council (PAC) regional reporting units.

We developed a single Bayesian Belief Network (BBN) habitat model (Fig. 2) with Netica v.1.17 software (Norsys Software Corp.) to analyze the historical and current conditions, and the effects of the SDEIS alternatives on elk and deer habitat capability. The model determined habitat capability within a subwatershed based on probability structures of the estimated influences of habitat variables (Marcot et al., 2001). Values of input variables and habitat capability outcomes were measured categorically at each node (low, moderate and high). Input values were assigned to the category that described the dominant condition in the subwatershed (Fig. 2). Relationships between variables (nodes) were described in conditional probability tables (CPT) based on the literature, or on expert opinion among the team members where data were lacking. An example

definition within a CPT was: "If road density is high, terrain is flat, and cover is low in a subwatershed, then elk security, expressed as the likely probability of each outcome category, is: low security = 0.90, moderate security = 0.10, and high security = 0.00". Similar probabilities of security were defined for all the other combinations of road density, terrain, and cover categories. The outcomes of the model in terms of habitat capability were represented by an expected value (similar to the mean), ranging from 1 to 3 based on integer ranking of the three outcome categories (low, moderate and high) and its standard deviation. The probability distribution of outcome categories also expressed the uncertainty associated with predicted outcomes. As opposed to expert panel evaluations that have been used to predict population viability outcomes (FEMAT, 1993; Lehmkuhl et al., 1997),

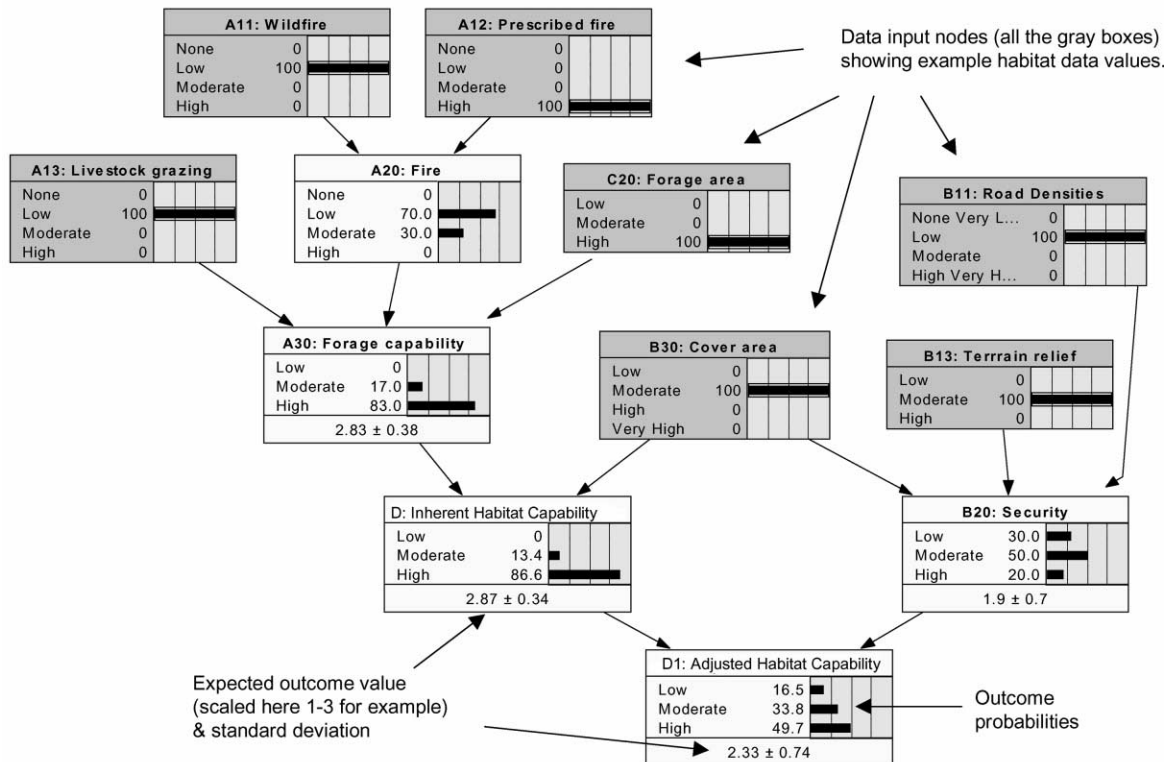


Fig. 2. Generic elk and deer BBN model for analysis of SDEIS alternatives for the ICBEMP. Diagram shows input data nodes (gray-tone) with example input values, and the calculated expected outcomes and outcome probabilities for those values.

our model quantified and structured expert and other information on habitat relationships to allow repeatable estimates of effects similar to conventional simulation models.

Our BBN model was developed to be consistent with the habitat correlates described by Christensen et al. (1995) and with the data that was available from the Project (Hemstrom et al., 2000). As such, the model was a balance between: (a) what is known or hypothesized about the habitat relationships of the species; (b) the type, resolution, and scale of vegetation data available across the Basin, as developed by the Project; (c) our ability to use those coarse data to model habitat relationships at the broad regional scale of the SDEIS analysis; and (d) the time available for the assignment (Marcot et al., 2001).

Lehmkuhl and Kie (2000) give a detailed description of the model variables and CPTs: a brief synopsis of model variables is given here. The model predicted both the inherent habitat capability and the adjusted

habitat capability of each subwatershed within the current range of the species. Inherent habitat capability was calculated as a function primarily of forage capability and secondarily cover area (Cook et al., 1998) within a subwatershed. Forage capability was defined primarily by the percentage area of forage habitat (early successional forest, herb, and shrub types among the 24 community types available — see Hemstrom et al., 2001), and the qualitative effects modeled by proxy variables. Proxy variables assessed the percentage area with uncharacteristic livestock grazing (negative effect from grazing levels causing >20% change in native plant community and soil structure), wildfire (positive effect by increasing herbaceous and shrub forage availability; also includes prescribed natural fire), and prescribed fire (positive effect similar to wildfire). Cover was defined as mid- and late-seral forest community types. Adjusted habitat capability was estimated by adjusting inherent habitat capability by the level of security from human

disturbance in the subwatershed. Security was determined by the interaction of cover area, road density, and topographic complexity (Edge and Marcum, 1991; Hillis et al., 1991).

Although the model structure (nodes and CPTs) was generic to all three species, model results varied for each species because forage and cover habitats were defined differently for each species, within the constraints of the broad cover types used for the SDEIS analysis. The BBN model can be run using the demonstration version of Netica (available at Norsys.com) using code available from the senior author (jlehmkuhl@fs.fed.us).

We estimated historical habitat capability as the inherent habitat capability based on vegetation conditions and disturbance regimes occurring about 150 years ago, as modeled by Hann et al. (1997). Those conditions did not reflect the effects of modern disturbance. Current and future habitat capabilities were estimated as the adjusted habitat capability using current and projected vegetation conditions, disturbance regimes, and road densities to adjust for the security effects of human disturbance. For each species and Region where the species occurred, we reported mean historical, current, and future habitat capabilities, weighted by subwatershed area, and the percentage of subwatersheds where future habitat capability was estimated to increase, decrease, or not change from current conditions. To map habitat capability by subwatershed, we categorized it as low (1.00–1.65), moderate (1.66–2.30), or high (2.31–3.0) based on breaks in the observed frequency distribution of values under current conditions. We defined a meaningful change in habitat capability to be  $\pm 0.5$  units, which approximated the average standard deviation (S.D.  $\approx 0.60$ ) in habitat capability of all 7467 subwatersheds for all time species, time periods, and alternatives.

### 2.3. Model sensitivity analysis

We did a sensitivity analysis of the BBN model to determine the relative influence of variables on habitat capability, hence, the underlying structure of the network relationships and the processes captured by the conditional probabilities (Marcot et al., 2001). We used the “entropy reduction” statistic generated by the sensitivity analysis procedure in the Netica

modeling software to quantify what percentage of the outcome at a particular node was attributed to each input variable, in a manner similar to reduction in variance described by the  $R^2$  statistic.

- Adjusted habitat capability was most sensitive to the level of inherent habitat capability (38% entropy reduction), with security having a small effect (6% entropy reduction).
- Inherent habitat capability was mostly a function of forage capability (67% entropy reduction), whereas cover had relatively little influence (2% entropy reduction).
- Security from human disturbance was most sensitive to changes in road density (29% entropy reduction), with lesser influences of cover (5% entropy reduction) and terrain relief (1% entropy reduction).
- Forage capability was primarily a function of forage area in a subwatershed (43% entropy reduction), with much weaker sensitivity to uncharacteristic livestock grazing (2% entropy reduction), wildfire (1% entropy reduction), and prescribed fire (0.02% entropy reduction).

## 3. Status of elk and deer

### 3.1. Historical habitats and populations

Elk appeared to have been widely distributed, although not always abundant, across most of the Basin during the early to mid-1800s before European settlement (Murie, 1951). Elk historically occurred in low numbers in the heavily forested portions of northern Idaho, northwestern Montana, and northeastern Washington (Murie, 1951), but were more numerous in the montane foothill regions of those areas, as well as in central and eastern Idaho, and northeastern Oregon. Elk were thought to have been historically absent from the central arid portions of eastern Washington and Oregon, and southwestern Idaho (Bailey, 1936; Bryant and Maser, 1982; Dalquist, 1948; Murie, 1951), where they are currently absent except for reintroduced populations or recent colonizers. However, McCorquodale (1985) and Dixon and Lyman (1996) showed from archeological data that elk were present in eastern Washington shrub-steppe habitats prior to European settlement. A few museum

records of elk in southeastern Oregon exist to indicate the recent historical presence of elk there (Murie, 1951). Our model estimated high historical habitat capability within the 79% of the Basin where elk currently occur (Table 1, Fig. 3A).

Mule deer historically were widespread in all habitats throughout the Basin, and their current distribution is unchanged. White-tailed deer historically were found in the northeastern portion of the Basin to the north and east of the arid interior. Compared with distribution records from 35 years ago (Ingles, 1965), Project range maps (on file at ICBEMP, 304 N. 8th, Boise, ID 83702) show white-tailed deer range expanding across northern Washington and northeastern Oregon. The current range of white-tailed deer covers 53% of the Basin. Within their current ranges across the Basin, our models showed historical mule deer and white-tailed deer habitat capability was high (Table 1, Figs. 4A and 5A).

Populations of all three species were estimated to have declined from the mid-1800s through the early 1900s during the period of Euro-American settlement. That decline is believed to be associated with habitat conversion to agriculture, widespread uncontrolled livestock grazing, and subsistence, market, or uncontrolled sport hunting (Murie, 1951; Schmidt, 1978). By the turn of the 20th century, the need for management was recognized (Schmidt, 1978). By the 1930s, elk and deer had begun a slow recovery through deliberate efforts to regulate hunting, preserve winter ranges, control competitive livestock grazing, and restock areas (mostly elk). In addition, from about 1900 through the 1930s, widespread wildfires created large areas of early successional habitats in northern and central Idaho, northeastern Washington, and western Montana, leading to abundant forage and high numbers of elk and mule deer in these areas well into the 1960s. Also, extensive clear-cut harvesting of timber from the 1950s through the 1980s created a mosaic of early successional foraging and cover habitats that have been a factor in increasing elk and white-tailed deer numbers in portions of the Basin.

Data on population levels were not consistently available region-wide for this analysis, but state records show that during the last 25 years, elk and mule deer numbers have varied across the Basin from stable to declining, depending on habitat and population dynamics and management policies. For example,

elk populations in Idaho's Lochsa River basin (Upper Columbia Salmon–Clearwater RAC) initially increased in response to improved habitat conditions resulting from wildfires in 1910, 1919, 1929, and 1934 that burned 71% of the basin. Subsequent population declines appeared to have been driven more by hunter harvest and extreme winter conditions during 1935, 1949, 1957, and 1976 than to the broad habitat changes following earlier fires. Observed elk population declines in the mid-1960s, initially believed to be the result of declining winter range conditions, were subsequently attributed to overharvest. When restrictions limiting harvest to bulls only were implemented in 1976, elk populations subsequently increased with little corresponding habitat enhancement.

Small, but rapidly expanding, elk populations occur in northeastern and north central Washington, and recolonization of shrub–steppe habitat in south-central Washington has established a vigorous new population there during the last 25 years (Eberhardt et al., 1996; Rickard et al., 1977). There is concern, however, that elk populations and productivity have been declining region-wide since the 1960s in many former strongholds in Oregon, Washington, and portions of Idaho as a result of both habitat change and population management policies. White-tailed deer numbers have been increasing in the last few decades, particularly in northern Idaho, northwestern Montana, eastern Washington, the Blue Mountains of Washington and Oregon, and the Snake River drainages of Oregon, Idaho, and Washington as a likely result of beneficial habitat changes, their high adaptability to coexistence with humans, and colonization of additional areas.

### 3.2. *Current habitat capability*

Elk habitat capability was estimated to have declined an average 35% across the Basin from historical levels (Table 1, Fig. 3). The 35% reduction included a 19% decline in inherent capability, plus a 16% negative security adjustment. If security effects were ignored, however, inherent habitat capability remained relatively high (mean = 2.3 vs. 2.8 for historical period), with 60% of the 15 Regions with high (2.31–3.00) capability, and only one Region with low (1.00–1.65) capability.

Changes in mule deer habitat from historical conditions followed a similar pattern. Habitat capability

Table 1

Bayesian model estimates of mean historical and current habitat capability for elk, mule deer, and white-tailed deer in the interior Columbia River basin, USA, and the estimated percentage changes in future habitat capability after 100 years under three ecosystem management alternatives<sup>a</sup>

| RAC/PAC (% RAC/PAC in species range)             | Historical<br>inherent | Current<br>adjusted | % change | % change (from current) |               |               |
|--------------------------------------------------|------------------------|---------------------|----------|-------------------------|---------------|---------------|
|                                                  |                        |                     |          | Alternative 1           | Alternative 2 | Alternative 3 |
| <i>Elk</i>                                       |                        |                     |          |                         |               |               |
| Sierra Front–Northwestern Great basin RAC (22%)  | 2.89                   | 2.03                | −29.7    | 0.0                     | 0.0           | 0.0           |
| Wyoming RAC (100%)                               | 2.35                   | 2.34                | −0.3     | 0.4                     | 0.4           | 0.4           |
| Lewiston RAC (100%)                              | 2.66                   | 1.82                | −31.5    | 2.3                     | 4.0           | −1.2          |
| Butte RAC (100%)                                 | 2.64                   | 1.59                | −39.7    | 14.3                    | 14.8          | 14.3          |
| Klamath PAC (93%)                                | 2.83                   | 1.78                | −37.0    | 2.4                     | 3.6           | 3.8           |
| Deschutes PAC (99%)                              | 2.79                   | 1.78                | −36.3    | 0.6                     | 1.1           | 0.9           |
| John Day–Snake RAC (72%)                         | 2.86                   | 1.65                | −42.4    | 9.4                     | 8.9           | 8.9           |
| Southeastern Oregon RAC (97%)                    | 2.88                   | 1.85                | −35.8    | 1.8                     | 2.0           | 2.0           |
| Lower Snake River RAC (51%)                      | 2.88                   | 1.89                | −34.5    | 1.3                     | 1.8           | 2.0           |
| Upper Snake River RAC (49%)                      | 2.84                   | 1.92                | −32.4    | 2.5                     | 2.8           | 2.9           |
| Upper Columbia Salmon–Clearwater RAC — R4 (97%)  | 2.74                   | 2.23                | −18.6    | 7.8                     | 7.7           | 7.6           |
| Upper Columbia Salmon–Clearwater RAC — R1 (99%)  | 2.78                   | 1.63                | −41.3    | 4.5                     | 4.4           | 4.0           |
| Eastern Washington RAC (31%)                     | 2.69                   | 1.42                | −47.3    | 21.2                    | 21.4          | 21.0          |
| Yakima PAC (97%)                                 | 2.72                   | 1.65                | −39.5    | 6.8                     | 6.8           | 6.8           |
| Eastern Washington Cascades PAC (24%)            | 2.41                   | 2.13                | −11.7    | −7.3                    | −7.3          | −7.3          |
| Weighted mean                                    | 2.77                   | 1.79                | −35.2    | 6.2                     | 6.4           | 6.2           |
| <i>Mule deer</i>                                 |                        |                     |          |                         |               |               |
| Sierra Front–Northwestern Great basin RAC (100%) | 2.99                   | 2.67                | −10.7    | −0.3                    | −0.2          | −0.2          |
| Wyoming RAC (100%)                               | 2.43                   | 2.37                | −2.3     | 3.2                     | 3.7           | 3.7           |
| Lewiston RAC (100%)                              | 2.70                   | 1.82                | −32.5    | 1.4                     | 4.9           | −0.3          |
| Butte RAC (100%)                                 | 2.64                   | 1.60                | −39.6    | 15.3                    | 15.5          | 14.9          |
| Klamath PAC (100%)                               | 2.84                   | 1.81                | −36.3    | 3.1                     | 4.0           | 4.1           |
| Deschutes PAC (100%)                             | 2.87                   | 1.94                | −32.2    | −1.1                    | −0.9          | −0.9          |
| John Day–Snake RAC (100%)                        | 2.89                   | 1.71                | −40.9    | 5.6                     | 5.6           | 5.5           |
| Southeastern Oregon RAC (100%)                   | 2.98                   | 2.31                | −22.6    | −3.8                    | −3.3          | −3.4          |
| Lower Snake River RAC (100%)                     | 2.97                   | 2.25                | −24.3    | −0.8                    | 0.0           | −0.1          |
| Upper Snake River RAC (100%)                     | 2.95                   | 2.03                | −31.3    | 3.0                     | 3.7           | 3.7           |
| Upper Columbia Salmon–Clearwater RAC — R4 (100%) | 2.77                   | 2.32                | −16.1    | 9.0                     | 9.6           | 9.4           |
| Upper Columbia Salmon–Clearwater RAC — R1 (100%) | 2.78                   | 1.63                | −41.4    | 3.9                     | 4.8           | 4.4           |
| Eastern Washington RAC (100%)                    | 2.87                   | 1.57                | −45.1    | 8.6                     | 8.9           | 8.8           |
| Yakima PAC (100%)                                | 2.81                   | 1.83                | −34.9    | 1.2                     | 1.4           | 1.4           |
| Eastern Washington Cascades PAC (100%)           | 2.55                   | 2.20                | −13.7    | −3.6                    | −2.7          | −2.8          |
| Weighted mean                                    | 2.85                   | 1.95                | −31.6    | 4.0                     | 4.5           | 4.3           |
| <i>White-tailed deer</i>                         |                        |                     |          |                         |               |               |
| Wyoming RAC (24%)                                | 2.65                   | 2.53                | −4.5     | −4.1                    | −2.1          | −2.1          |
| Lewiston RAC (100%)                              | 2.56                   | 1.82                | −28.8    | −0.7                    | 2.7           | −2.4          |
| Butte RAC (100%)                                 | 2.63                   | 1.59                | −39.5    | 12.8                    | 13.0          | 12.4          |
| John Day–Snake RAC (40%)                         | 2.88                   | 1.55                | −46.0    | 2.2                     | 1.4           | 1.2           |
| Lower Snake River RAC (25%)                      | 2.93                   | 1.72                | −41.4    | 2.6                     | 2.2           | 2.4           |
| Upper Snake River RAC (74%)                      | 2.91                   | 2.04                | −29.6    | 0.1                     | 0.6           | 0.7           |
| Upper Columbia Salmon–Clearwater RAC — R4 (90%)  | 2.65                   | 2.33                | −12.0    | 3.9                     | 4.4           | 4.7           |
| Upper Columbia Salmon–Clearwater RAC — R1 (100%) | 2.78                   | 1.63                | −41.4    | 2.9                     | 3.7           | 3.4           |
| Eastern Washington RAC (59%)                     | 2.78                   | 1.42                | −49.1    | 11.3                    | 11.8          | 11.6          |
| Eastern Washington Cascades PAC (38%)            | 2.43                   | 2.07                | −14.8    | −6.8                    | −3.5          | −5.3          |
| Weighted mean                                    | 2.76                   | 1.79                | −35.1    | 5.3                     | 5.7           | 5.5           |

<sup>a</sup> Inherent capability was a function of forage and cover quantity and quality. Adjusted capability for current and alternative futures was inherent capability adjusted for the level of security from human disturbance.

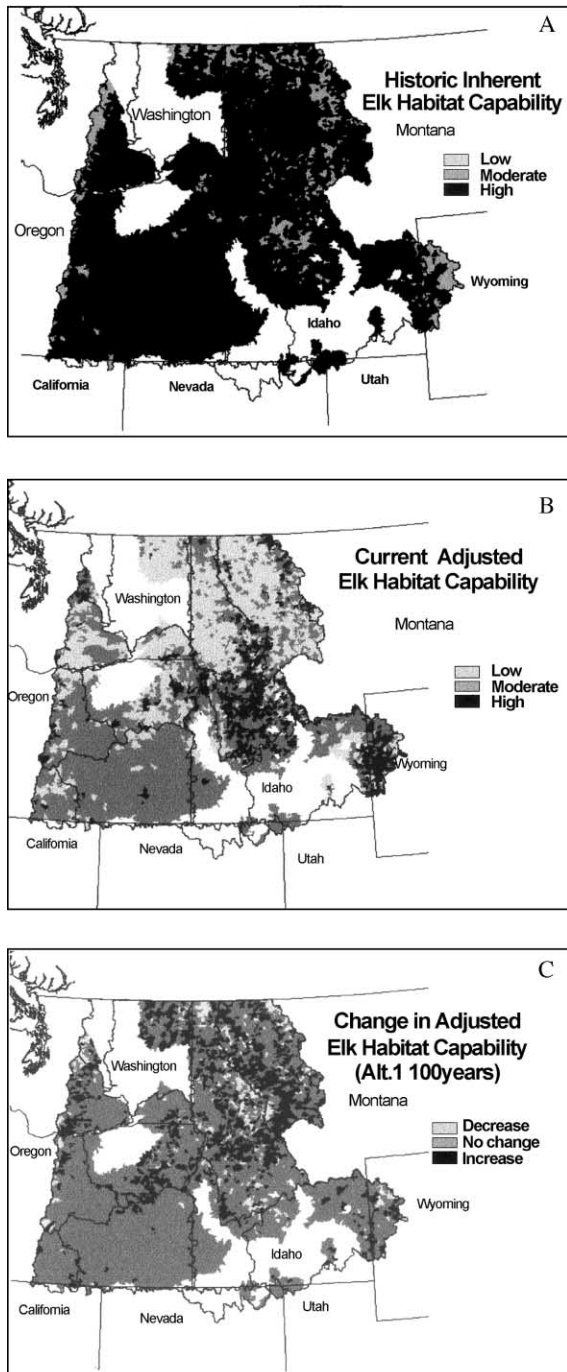


Fig. 3. Historical elk habitat capability (A), current habitat capability adjusted for security from human disturbance (B), and projected changes in habitat capability after 100 years under ecosystem management Alternative 1 (C) of the SDEIS for the ICBEMP.

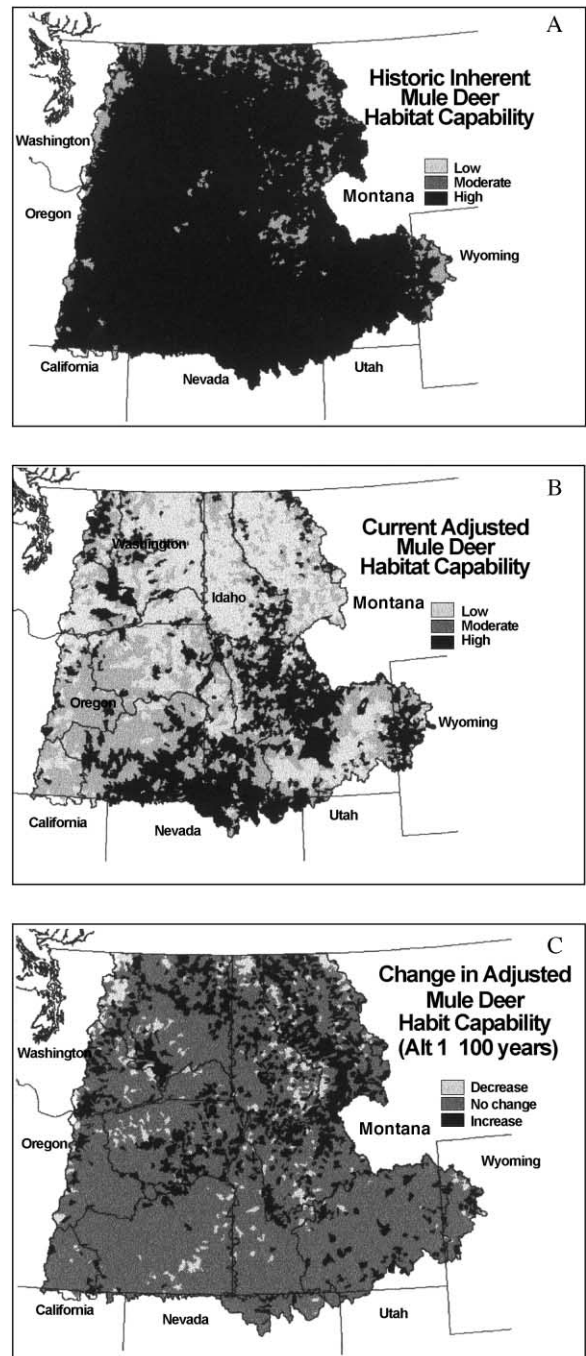


Fig. 4. Historical mule deer habitat capability (A), current habitat capability adjusted for security from human disturbance (B), and projected changes in habitat capability after 100 years under ecosystem management Alternative 1 (C) of the SDEIS for the ICBEMP.

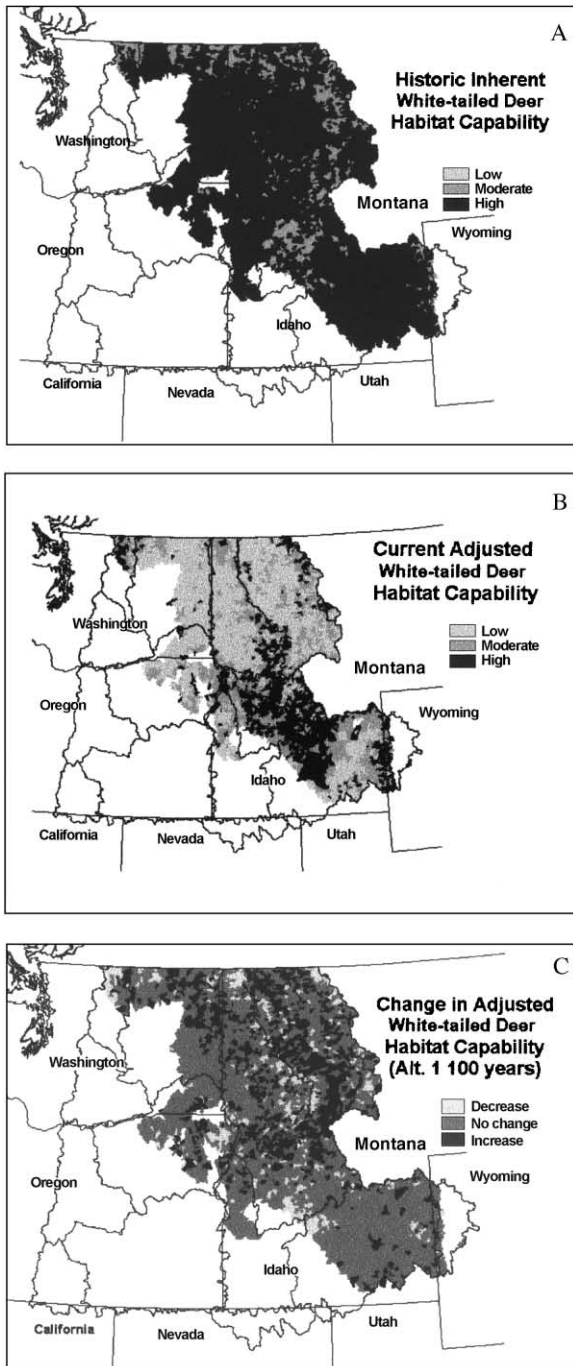


Fig. 5. Historical white-tailed deer habitat capability (A), current habitat capability adjusted for security from human disturbance (B), and projected changes in habitat capability after 100 years under ecosystem management Alternative 1 (C) of the SDEIS for the ICBEMP.

was estimated to have declined 32% across the Basin compared to historical values (Table 1, Fig. 4). Thirteen percent of the Regions scored high, 30% scored moderately high (2.0–2.3), and the rest moderate to moderately low ( $>2.0$ ) (Table 1). About 19% of the decline was attributed to a drop in inherent capability, whereas loss of security accounted for the other 13%. Discounting the security adjustment, however, inherent habitat capability for mule deer remained high across the Basin (mean = 2.3 vs. 2.9 for historical period), with high inherent capability in 60% of 15 Regions, and moderate inherent capability in the others.

White-tailed deer habitat changed in a fashion similar to elk and mule deer. Habitat capability was estimated to have declined 35% across the Basin compared to historical values (Table 1, Fig. 5). Only two Regions scored within the high range of habitat capability, five scored moderately high (2.0–2.3), and the rest were moderate to moderately low ( $>1.6$ ). About 25% of the decline was from a drop in inherent habitat capability, whereas loss of security accounted for the other 10%. Ignoring the security adjustment, inherent habitat capability for white-tailed deer was moderate across the Basin (mean = 2.1 vs. 2.8 for historical period), but locally high in 45% of the 11 Regions with white-tailed deer.

#### 4. Effects of future management

##### 4.1. Key assumptions

Assumptions about institutional and general process issues, and assumptions to ensure consistency among the various science evaluations are described by Quigley (2000). Assumptions for modeling and predicting changes over time in the habitat variables (e.g., terrestrial community types that described forage and cover area) and disturbance variables (uncharacteristic livestock grazing, wildfire, prescribed fire) are described by Hann et al. (1997) and Hemstrom et al. (2000). Our additional assumptions for modeling elk and deer habitat capability were:

- The structure and probability tables of the BBN model estimate habitat capability equally well for elk, mule deer, and white-tailed deer.

- Habitat capability indicates the relative carrying capacity for ungulates, but does not predict actual population levels or productivity, which are affected by abiotic and human factors that were not modeled.
- Proxy variables in the SDEIS dataset that influenced forage capability (wildfire, prescribed fire, uncharacteristic livestock grazing) accurately estimated the qualitative effects of fire and grazing on forage capability.
- Projected changes in road density are from the obliteration of roads, and did not include temporary road closures by gates and other means that might improve elk and deer security from human disturbance.

#### 4.2. Future trends

Elk habitat capability across the Basin was projected to increase about 6% from the current situation over a 100-year period under all three SDEIS alternatives (Table 1). Two Regions showed larger than average increases of 14% (Butte) and 21% (eastern Washington). Security would decrease about 15% relative to current conditions, so change was primarily from increased forage capability. In the short term (10 years), elk habitat capability under all alternatives would increase in 9% of the subwatersheds that make up individual Regions, decline in 1%, and remain unchanged in the remaining 90%. After 100 years, however, habitat capability would increase in about 20% of the subwatersheds and declined in 6% (Fig. 3C). Projected increases in habitat capability would be well distributed across the mountainous elk range in the Basin. Most increases would occur in western Montana, eastern and northeastern Washington, portions of northern and central Idaho, and northeastern Oregon. The largest declines in capability over the next 100 years were projected to occur in the eastern Washington Cascades, and portions of northern and central Idaho.

Mule deer habitat capability was estimated to increase about 4% across the Basin under all ecosystem management alternatives over the next 100 years (Table 1). Security would decrease about 17%, so most of the change would be from better forage and cover conditions. In the short term, habitat capability would increase in about 9% of the subwatersheds under all

alternatives, decline in 1%, and remain unchanged in 90%. After 100 years, though, mule deer habitat capability was projected to increase similarly among alternatives in about 18% of the subwatersheds, while 6% would decline in capability (Fig. 4C). Projected increases would be well distributed across the mountainous mule deer range in the Basin, primarily where habitat capability is currently rated low. Most increases would occur in western Montana, throughout eastern Washington, in northern and central Idaho, and northeastern Oregon. The largest declines in capability over the next 100 years were projected to occur in the eastern Washington Cascades and portions of northern Idaho.

White-tailed deer habitat capability was projected to increase about 5–6% from the current situation over the next 100 years under all ecosystem management alternatives (Table 1). Security would decrease about 15%, so most changes were from better forage and cover conditions. In the short term, habitat capability would increase in about 9–10% of the subwatersheds, decline in 3–4%, and remain unchanged in 85–88%. After 100 years, all alternatives were estimated to increase white-tailed deer habitat capability in 18% of the subwatersheds, while 6% would decline (Fig. 5C). Projected increases would be well distributed across white-tailed deer range in the northern Rocky Mountains and northeastern Washington, primarily where habitat capability is currently rated low. Most increases would occur in western Montana, northeastern Washington, in portions of northern Idaho in the Bitterroot Range, and in central Idaho. The largest declines in capability over the next 100 years were projected to occur in the eastern Washington Cascades, and in portions of northern Idaho, including the Bitterroot Range.

## 5. Discussion

### 5.1. Drivers of change

Historical declines in habitat capability were a result of Basin-wide shifts in vegetation communities that lowered forage capability, plus the effects of reduced security from roads. Using the same Project dataset as our analysis, Hann et al. (1997) estimated that early seral lower montane and mid-montane forest

foraging habitats declined 77 and 8% with succession to closed-canopy mid-seral stages. Upland herb and shrub foraging habitats decreased 67 and 31%, respectively, with the primary transition being to agriculture and exotic species.

Ecosystem management alternatives were projected to maintain or slightly increase habitat capability for elk and deer across the Basin. Although the average projected increase in habitat capability across the Basin was relatively low (<5%), habitat capability in some areas dominated by federal landholdings and management showed greater predicted increases (Figs. 3–5). Based on results of broad-scale vegetation modeling of SDEIS alternatives, the slight increases we predicted in habitat capability appeared to be driven by the qualitative effects on forage of fire and grazing management, and, perhaps, by some localized decreases in road density.

Vegetation modeling for the SDEIS analysis estimated that the area of forage habitats mostly would decline somewhat or remain little changed (Hemstrom et al., 2001). Early seral forest forage types were projected to decline due to succession and decreased timber harvest, while non-forage mid-seral forests would continue to dominate landscapes. Upland shrub and herb forage types showed some small gains on federal lands, while herbland dominated by non-forage exotic species showed some decline on federal lands. Beneficial qualitative influences on modeled forage capability were estimated to increase, however. Although wildfire area would decrease slightly, prescribed fire and fuel management (thinning) would increase about 60% on federal lands. Uncharacteristic livestock grazing (causing shifts in vegetation community and soil structures) would decline under all alternatives.

Ungulate security would change little over most of the Basin under the alternatives, but might improve in a few areas. Permanent road closures (obliteration) over the next 100 years would be relatively small overall, affecting about 9% of the Basin (Hemstrom et al., 2000). Road closures would be concentrated in grizzly bear (*Ursus arctos*) recovery areas, thus possibly contributing to projected increases in habitat capability for those areas, primarily the northern Rocky Mountains of Montana and Idaho and in the eastern Washington Cascade Range. However, based on sensitivity analysis the modeled relationship

between habitat capability and security was weak and results often showed an inverse relationship between changes in habitat capability and security from current values; i.e., areas with the greatest increase in habitat capability had the greatest loss in security. For example, elk habitat capability increased about 14% in the Butte RAC, an important part of the grizzly bear recovery zone; but, security there decreased 64% from current conditions under the alternatives. Likewise, elk habitat capability in the eastern Washington RAC increased 21% while security decreased 114% from current.

Thus, changes in habitat capability appeared to be a function of changes in forage capability, while concurrent decreases in cover might have contributed to less security. Although percentage decreases in security from the current period appeared to be high in some cases, the change was nevertheless small in absolute terms — less than 0.5 units of capability in all cases. Road density likely would have had a stronger influence on habitat capability, as modeled, if more roads had been permanently closed, or if temporary road closures (gates, berms and trenches) commonly used to limit human access to critical areas were considered in the overall SDEIS analysis.

## 5.2. Scale and model complexity

Our model was designed for a coarse regional-scale analysis of relative changes in habitat capability among alternatives. That scale imposed limitations on the available data, the structure of the model, and the interpretation and application of results at smaller scales. The necessary mapping of vegetation as the dominant terrestrial community type at a 1 km<sup>2</sup> pixel resolution had important implications (Hann et al., 1997). Terrestrial community types were broad categories of vegetation (e.g., “early seral lower montane forest”) that lumped a variety of cover and structural stages, thus masking smaller-scale variation in forage or cover habitats. For example, a pixel dominated by 60% closed canopy forest with 40% meadow or herbaceous vegetation would have been classified as cover, thus reducing the forage capability for that pixel. Thus, the area of an individual community type in a subwatershed, calculated as the sum of cell values, may have mapping error. However, that error declines when similar community types

are combined, as we did to estimate forage and cover area, and with increasing size of the reporting unit, with the lowest error rate at the regional scale (Hann et al., 1997) as reported here.

The coarse resolution of mapping by dominant type within a map pixel also systematically underestimated the area of a few potentially important habitats, such as riparian vegetation, because their distribution was finer than the mapping resolution (Hann et al., 1997). Hence, habitat capability in some areas might be underestimated for species for which riparian areas are important habitat, such as white-tailed deer (Christensen et al., 1995). Coarse mapping resolution also could not quantify the fine-scale spatial pattern of cover and forage habitats that might be an important element of habitat quality (Thomas et al., 1988; Wisdom et al., 1986); however, that fine-scale pattern of cover and forage might be less important than previously thought (Cook et al., 1998).

Finally, habitat types were mapped too coarsely to include complete information on forage quality — a critical component of habitat capability that determines herd productivity and population fitness (Hobbs and Swift, 1985). The coarse vegetation types used for mapping habitats include many plant associations with different levels of inherent forage quality. Proxy variables available to describe qualitative influences on forage capability (uncharacteristic livestock grazing, wildfire, and prescribed fire levels) likely were incomplete for comprehensive modeling of qualitative variation in forage capability within vegetation types. For example, elk and deer browsing can alter plant communities every bit as much as livestock, but only information on livestock grazing was available. Similarly, high fire frequency can increase nutrient loads of forages, but this is of use only if palatable forages have not been eliminated by past use. It is doubtful, however, that such fine-scale information could ever be systematically collected and made available for analyses of habitat capability at regional scales.

### 5.3. Model validation

Validation of habitat capability models is inherently a difficult task (Laymon and Barrett, 1986; Roloff and Kernohan, 1999). Although habitat capability has an intuitive definition, defining it in ways that can be

measured for validation is difficult. Ideally, habitat capability indexes carrying capacity, and associated population size, or productivity measured as pregnancy rates or calf survivorship, such that we might expect a correlation. We have not made that claim for this model for the well-known reason that habitat is not always a good measure of population response (Van Horne, 1983). For example, as described earlier, Lochsa elk population size and productivity were found to be not merely a function of habitat capability, but also of management as it affects other influences such as hunter harvest. Management also may affect sex and age ratios and predation rates that influence population size or productivity.

Density-dependent factors for which data are not available Basin-wide also may affect the habitat-population correlations (Roloff and Kernohan, 1999). For example, the Wenaha wilderness area of the Blue Mountains (northern part of John Day–Snake RAC) was rated relatively high in habitat capability by our model, but that area has the least productive elk subpopulation in the Blue Mountains. The area is rated high presumably because the absence of roads and presence of steep topography result in excellent security, allowing the high inherent capability to be expressed. Unfortunately, elk currently do not do well there. One possible reason is that elk numbers peaked in the 1970s, and have declined as a result of self-induced habitat change. Based on observations in exclosures, high-density populations throughout the Blue Mountains during the 1970s likely eliminated preferred high-quality forage, particularly in the Wenaha. Although early successional forage types are abundant, the quality of forage is poor and hence productivity low.

The alternative, thus, is to validate habitat capability without comparison to population data. That procedure necessarily would involve expert opinion of habitat capability — an apparent tautology. That exercise also would insert an additional element of subjectivity and variation associated with differing and perhaps unrepeatable expert opinions on what constituted habitat capability, a situation that we attempted to avoid by developing this model.

Given the difficulties of empirically validating habitat capability and the lack of consistent and readily available regional data on ungulate populations, we suggest that the “validity” of this model be judged by

different criteria. One criterion would be the way it structured available knowledge of habitat relationships and linked Project-specific habitat variables to address the assigned question of how large-scale management proposals would affect ungulate habitats. We believe the model fit that criterion.

#### 5.4. Conclusions

Our model structured broad-scale vegetation and disturbance variables from an ecosystem management planning dataset into a BBN model to estimate changes in elk and deer habitat capability among land management planning alternatives. The model estimated that the ecosystem management alternatives would equally maintain or slightly improve elk and deer habitat capability in most of the 15 Regions in the Basin. While useful for regional analyses, the model may be too coarse for determining changes in smaller areas that are traditionally analyzed for ungulate habitat capability. Constraints on regional mapping of vegetation resulted in coarse definitions of forage, excluded small-grain habitats, or could not completely quantify qualitative factors. Nevertheless, the model did structure available knowledge of habitat relationships and linked Project-specific habitat variables to address the assigned question of how large-scale management proposals would affect ungulate habitats.

Analysis of elk and deer habitat capability at smaller subwatershed scales useful for forest planning and localized management would be most appropriately done with data acquired at a resolution higher than 1 km<sup>2</sup>. For example, subwatershed vegetation mapping at a 4 ha resolution from aerial photographs and the associated analysis of disturbance regimes by Hessburg et al. (1999) for ICBEMP would be the appropriate scale for subwatershed or smaller-scale habitat analyses. Also, forage quality and security issues related to permanent and temporary road closures could be more adequately addressed at a subwatershed scale. Models of that scale might be more easily linked with population dynamics.

#### Acknowledgements

Model development was a collaborative effort of other scientists and managers on the analysis team:

Craig Bienz and Don Gentry (Klamath Tribes), Mark Kirsch and Scott Findholt (Oregon Department of Fish and Wildlife), Scott McCorquodale (Yakama Indian Nation), and Mike Hillis (Lolo National Forest, Montana). Bruce Marcot provided critical instruction and comments on the development and use of BBN models.

#### References

- Bailey, V., 1936. The mammals and life zones of Oregon, North American Fauna Edition. US Department of Agriculture, Bureau of Biological Survey, Washington, DC.
- Bettinger, P., Johnson, K.N., Sessions, J., 1996. Forest planning in an Oregon case study: defining the problem and attempting to meet goals with spatial-analysis technique. *Environ. Manage.* 20, 565–577.
- Bryant, L.D., Maser, C., 1982. Classification and distribution. In: Thomas, J.W., Toweill, D.E. (Eds.), *Elk of North America: Ecology and Management*. Stackpole Books, Harrisburg, PA, pp. 1–60.
- Christensen, A.G., Lyon, J.L., Pederson, R., Bodie, W.L., Johnson, R., O'Gara, B., 1995. Ungulate assessment (sic) in the Columbia River basin. Interior Columbia Basin Ecosystem Management Project. USDA Forest Service, US Department of Interior, Bureau of Land Management, Boise, ID.
- Cook, J.G., Irwin, L.L., Bryant, L.D., Riggs, R.A., Thomas, J.W., 1998. Relations of forest cover and condition of elk: a test of the thermal cover hypothesis in summer and winter. *Wildl. Monogr.* 141, 1–61.
- Dalquist, W.W., 1948. *Mammals of Washington*, Museum of Natural History, University of Kansas.
- Dixon, S.L., Lyman, R.L., 1996. On the Holocene history of elk (*Cervus elaphus*) in eastern Washington. *Northw. Sci.* 70, 262–272.
- Dombeck, M.P., 1996. Thinking like mountain: BLM's approach to ecosystem management. *Ecol. Appl.* 6, 699–702.
- Eberhardt, L.E., Eberhardt, L.L., Tiller, B.L., Cadwell, L.L., 1996. Growth of an isolated elk population. *J. Wildl. Manage.* 60, 369–373.
- Edge, W.D., Marcum, C.L., 1991. Topography ameliorates the effects of roads and humans. In: Christensen, A.G., Lyon, J.L., Lonner, T.N. (Eds.), *Proceedings of the Elk Vulnerability Symposium*. Montana Department of Fish, Wildlife, and Parks, Bozeman, MT.
- FEMAT (Forest Ecosystem Management Assessment Team), 1993. *Forest ecosystem management: an ecological, economic, and social assessment*. US Department of Agriculture, US Department of the Interior, US Department of Commerce, and Environmental Protection, Portland, OR (irregular pagination).
- Hann, W.J., Karl, M.G.S., Jones, J.L., Gravenmier, R.A., Long, D.G., Menakis, J.P., Keane, R.E., 1997. Landscape ecology evaluation of the preliminary draft EIS alternatives. In: Quigley, T.M., Lee, K.M., Arbelbide, S.J. (Tech. Eds.), *Evaluation of*

- EIS Alternatives by the Science Integration Team, Vol. I. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, pp. 29–432.
- Haynes, R.W., Horne, A.L., 1997. Economic assessment of the Basin. In: Quigley, T.M., Arbelbide, S.J. (Tech. Eds.), *An Assessment of Ecosystem Components in the Interior Columbia River basin and Portions of the Klamath and Great Basins*, Vol. 4. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, pp. 1715–1869.
- Haynes, R.W., Quigley, T.M., Clifford, J.L., Gravenmier, R.A., 2001. Science and ecosystem management in the interior Columbia River basin. *For. Ecol. Manage.* 153, 3–14.
- Hemstrom, M.A., Hann, W.J., Gravenmier, R.A., Korol, J.J., 2000. Draft landscape effects analysis of the SDEIS alternatives. In: Science Advisory Group. Draft March 2000. Science Advisory Group Effects Analysis for the SDEIS Alternatives. Interior Columbia Basin Ecosystem Management Project. USDA Forest Service, US Department of Interior, Bureau of Land Management, Boise, ID.
- Hemstrom, M.A., Korol, J.J., Hann, W.J., 2001. Trends in terrestrial plant communities and landscape health indicate the effects of alternative management strategies in the interior Columbia River basin. *For. Ecol. Manage.* 153, 105–125.
- Hessburg, P.F., Smith, B.G., Kreiter, S.D., Miller, C.A., Salter, R.B., McNicoll, C.H., Hann, W. J., 1999. Historical and current forest and range landscapes in the interior Columbia River basin and portions of the Klamath and Great basins. Part 1. Linking vegetation patterns and landscape vulnerability to potential insect and pathogen disturbances. General Technical Report PNW-GTR-458. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, 357 pp.
- Hillis, J.M., Thompson, M.J., Canfield, J.E., Lyon, J.L., Marcum, C.L., Dolan, P.M., McCleerey, D.W., 1991. Defining elk security: the Hillis paradigm. In: Christensen, A.G., Lyon, J.L., Lonner, T.N. (Eds.), *Proceedings of the Elk Vulnerability Symposium*. Montana Department of Fish, Wildlife, and Parks, Bozeman, MT, pp. 38–43.
- Hobbs, N.T., Swift, D.M., 1985. Estimates of habitat at carrying capacity incorporating explicit nutritional constraints. *J. Wildl. Manage.* 49 (3), 814–822.
- Ingles, L.G., 1965. *Mammals of the Pacific States*. Stanford University Press, Stanford, CA.
- Jourdonnais, C.S., Bedunah, D.J., 1990. Prescribed fire and cattle grazing on an elk winter range in Montana. *Wildl. Soc. Bull.* 18, 232–240.
- Laymon, S.A., Barrett, R.H., 1986. Developing and testing habitat-capability models: pitfalls and recommendations. In: Verner, J., Morrison, M.L., Ralph, C.J. (Eds.), *Wildlife 2000: Modeling Habitat Relationships of Terrestrial Vertebrates*. University of Wisconsin Press, Madison, WI, pp. 87–91.
- Lehmkuhl, J.F., Kie, J., 2000. Current and projected habitat trends of elk, mule deer, and white-tailed deer under alternatives for the ICBEMP supplemental draft EIS (SDEIS). In: Science Advisory Group. Draft March 2000. Science Advisory Group Effects Analysis for the SDEIS Alternatives. Interior Columbia Basin Ecosystem Management Project. USDA Forest Service, US Department of Interior, Bureau of Land Management, Boise, ID.
- Lehmkuhl, J.F., Raphael, M.G., Holthausen, R.S., Hickenbottom, J.R., Naney, R.H., Shelly, J.S., 1997. Historical and current status of terrestrial species and the effects of proposed alternatives. In: Quigley, T.M., Lee, K.M., Arbelbide, S.J. (Tech. Eds.), *Evaluation of EIS Alternatives by the Science Integration Team*, Vol. II. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, pp. 537–730.
- Lyon, J.L., Ward, A.L., 1982. Elk and land management. In: Thomas, J.W., Toweill, D.E. (Eds.), *Elk of North America*. Stackpole Books, Harrisburg, PA, pp. 443–477.
- Marcot, B.G., Holthausen, R.S., Raphael, M.G., Rowland, M.M., Wisdom, M.J., 2001. Using Bayesian Belief Networks to evaluate wildlife population viability under land management alternatives from an environmental impact statement. *For. Ecol. Manage.* 153, 29–42.
- McCabe, R.E., 1982. Elk and Indians: historical values and perspectives. In: Thomas, J.W., Toweill, D.E. (Eds.), *Elk of North America*. Stackpole Books, Harrisburg, PA, pp. 61–124.
- McCool, S.F., Burchfield, J.A., Allen, S.D., 1997. Social assessment. In: Quigley, T.M., Arbelbide, S.J. (Tech. Eds.), *An assessment of ecosystem components in the interior Columbia River basin and portions of the Klamath and Great basins*, Vol. 4. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, pp. 1871–2009.
- McCorquodale, S.M., 1985. Archaeological evidence of elk in the Columbia basin. *Northw. Sci.* 59, 192–197.
- Murie, O.J., 1951. *Elk of North America*. Stackpole Books, Harrisburg, PA.
- Nelson, J.R., 1982. Relationships of elk and other large herbivores. In: Thomas, J.W., Toweill, D.E. (Eds.), *Elk of North America*. Stackpole Books, Harrisburg, PA, pp. 415–441.
- Quigley, T.M. (Tech. Ed.), 2000. Science Advisory Group. Draft March 2000. Science Advisory Group Effects Analysis for the SDEIS Alternatives. Interior Columbia Basin Ecosystem Management Project. USDA Forest Service, US Department of Interior, Bureau of Land Management, Boise, ID.
- Rickard, W.H., Hedlund, J.D., Fitzner, R.E., 1977. Elk in the shrub-steppe of Washington: an authentic record. *Science* 196, 1009–1010.
- Roloff, G.J., Kernohan, B.J., 1999. Evaluating reliability of habitat suitability index models. *Wildl. Soc. Bull.* 27, 973–985.
- Schmidt, J.L., 1978. Early management: intentional and otherwise. In: Schmidt, J.L., Gilbert, D.L. (Eds.), *Big Game of North America*. Stackpole Books, Harrisburg, PA, pp. 257–270.
- Steinhoff, H.W., 1978. Big game values. In: Schmidt, J.L., Gilbert, D.L. (Eds.), *Big Game of North America*. Stackpole Books, Harrisburg, PA, pp. 271–282.
- Thomas, J.W., Dombeck, M., 1996. Ecosystem management in the interior Columbia River basin. *Wildl. Soc. Bull.* 24, 180–186.
- Thomas, J.W., Black Jr., H., Scherzinger, R.J., Pedersen, R.J., 1979. Deer and elk. In: Thomas, J.W. (Ed.), *Wildlife Habitats in Managed Forests: The Blue Mountains of Oregon and Washington*. US Department of Agriculture, Forest Service, Washington, DC, pp. 104–127.

- Thomas, J.W., Leckenby, D.A., Henjum, M., Pedersen, R.J., Bryant, L.D., 1988. Habitat-effectiveness index for elk on Blue Mountain winter ranges. General Technical Report PNW-218. US Department of Agriculture, Forest Service, Pacific Northwest Research Station, Portland, OR, 28 pp.
- USDA and USDI, 2000. Interior Columbia basin supplemental draft environmental impact statement, BLM/OR/WA/Pt-00/019+1972. US Department of Agriculture, Forest Service, US Department of the Interior, Bureau of Land Management, Portland, OR (irregular pagination). <http://www.icbemp.gov>.
- Van Horne, B., 1983. Density as a misleading indicator of habitat quality. *J. Wildl. Manage.* 47, 893–901.
- Wisdom, M.J., Bright, L.R., Carey, C.G., Hines, W.W., Pedersen, R.J., Smithy, D.A., Thomas, J.W., Witmer, G.W., 1986. A model to evaluate elk habitat in western Oregon, Vol. 216. US Department of Agriculture, Forest Service, Pacific Northwest Region, and US Department of the Interior, Bureau of Land Management and Oregon Department of Fish and Wildlife, 2, 36 pp.